Prompt engineering

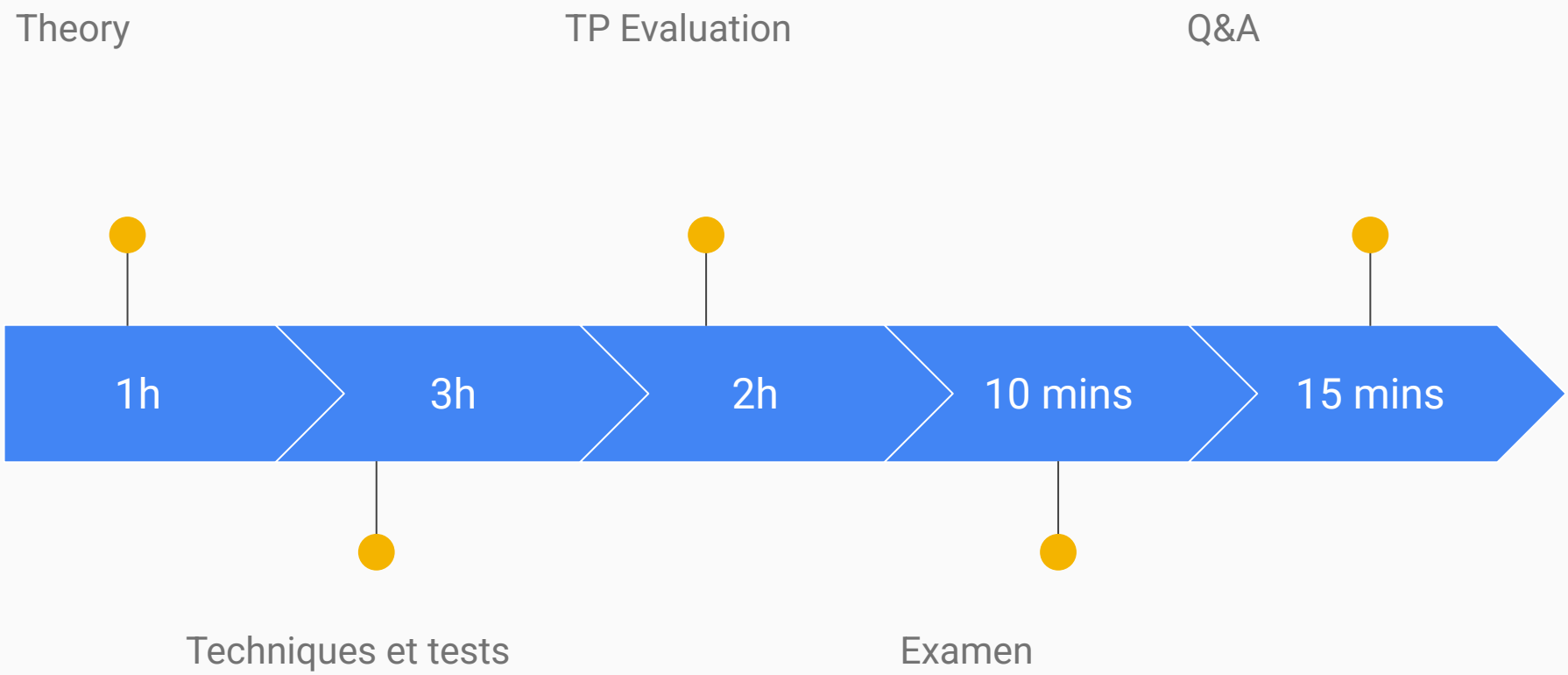
Objectif

Compréhension approfondie des principes et des meilleures pratiques associées au prompts engineering, essentiels pour exploiter efficacement les modèles d'intelligence artificielle générative, notamment dans la création de contenu, l'interact

naviguer efficacement dans le domaine en évolution rapide du prompt engineering, en mettant un accent particulier sur la manière de tirer le meilleur parti des interactions avec les systèmes d'IA générative pour des applications pratiques et créatives. Et en améliorant l'interaction homme-IA de manière éthique et responsable.

Materials

- Item 1
- Item 2
- Item 3
- Verification
- Qualification ?
- QCM


Theory
TP Evaluation
Q&A
1h
3h
2h
10 mins
15 mins
Techniques et tests
Examen

Theory

Theory

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs)

Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning.

Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Common settings 1

Temperature - the lower the temperature, the more deterministic the results : the highest probable next token is always picked. Increasing temperature could lead to more randomness,

Top P - A sampling technique with temperature. A high top_p value will enable the model to look at more possible words, including less likely ones, leading to more diverse outputs.

The general recommendation is to alter temperature or Top P but not both.

Max Length - You can manage the number of tokens the model generates by adjusting the max length. Specifying a max length helps you prevent long or irrelevant responses and control costs.

Stop Sequences - For example, you can tell the model to generate lists that have no more than 10 items by adding "11" as a stop sequence.

Common settings 1

Frequency Penalty - The frequency penalty applies a penalty on the next token proportional to how many times that token already appeared in the response and prompt. The higher the frequency penalty, the less likely a word will appear again.

Presence Penalty - Unlike the frequency penalty, the penalty is the same for all repeated tokens. This setting prevents the model from repeating phrases too often in its response. If you want the model to generate diverse or creative text, you might want to use a higher presence penalty. Or, if you need the model to stay focused, try using a lower presence penalty.

Similar to temperature and top_p, the general recommendation is to alter the frequency or presence penalty but not both.

Prompting

You can achieve a lot with simple prompts, but the quality of results depends on how much information you provide it and how well-crafted the prompt is.

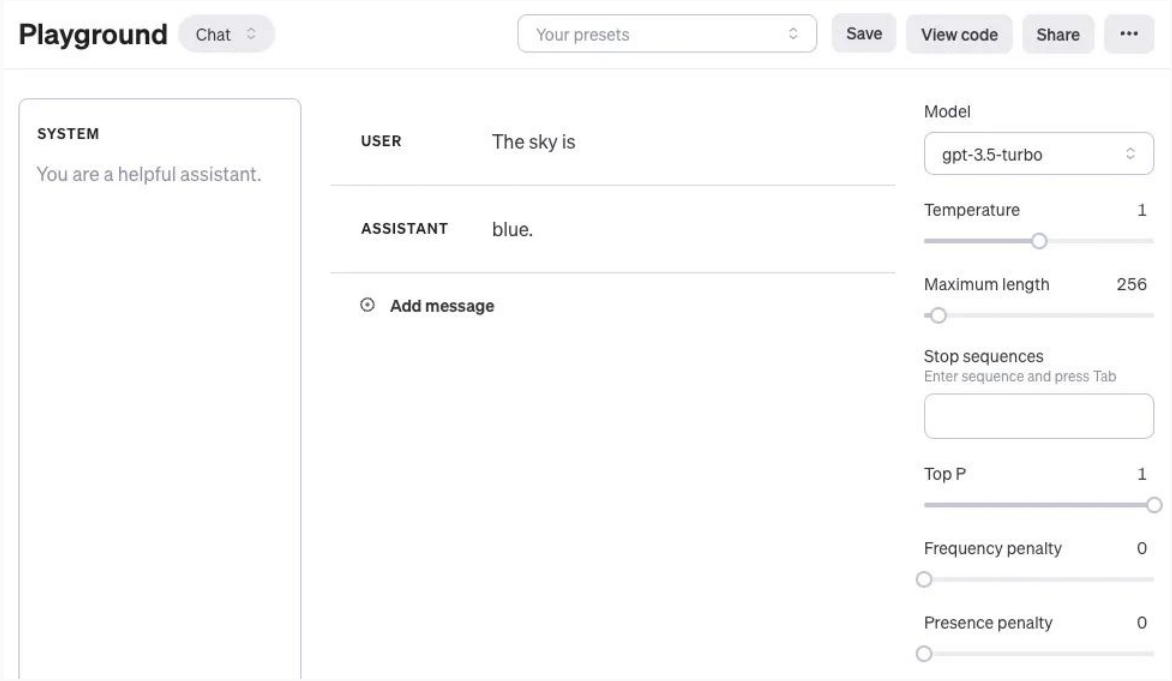
A prompt can contain information like

- the instruction
- question

And other details such as

- context,
- inputs
- examples.

Exemple


Playground
Chat
Your presets
Save
View code
Share
SYSTEM
You are a helpful assistant.
USER
The sky is
ASSISTANT
blue.
Add message
Model
gpt-3.5-turbo
Temperature
1
Maximum length
256
Stop sequences
Enter sequence and press Tab
Top P
1
Frequency penalty
0
Presence penalty
0

Limits

We can observe from the exemple

- the language model responds with a sequence of tokens that make sense given the context "The sky is".
- The output might be unexpected or far from the task you want to accomplish.

=> necessity to provide **more context or instructions on what specifically you want to achieve with the system.**

This is what prompt engineering is all about.

Let’s improve

Let's try to improve it a bit:

Prompt:

> Complete the sentence:
>
> The sky is

Output:

blue during the day and dark at night.

Is that better?

Well, with the prompt above you are instructing the model to complete the sentence so the result looks a lot better as it follows exactly what you told it to do ("complete the sentence").

Prompt Formatting

You have tried a very simple prompt above. A standard prompt has the following format:

<Question>?

or

<Instruction>

You can format this into a question answering (QA) format, which is standard in a lot of QA datasets, as follows:

Q: <Question>?

A:

Few-shot prompting

Given the standard format above, one popular and effective technique to prompting is referred to as **few-shot prompting** where you provide exemplars (i.e., demonstrations). You can format few-shot prompts as follows:

<Question>?
<Answer>
<Question>?
<Answer>
<Question>?
<Answer>
<Question>?

The QA format version would look like this:

Q: <Question>?
A: <Answer>
Q: <Question>?
A: <Answer>
Q: <Question>?
A: <Answer>
Q: <Question>?
A:

Example

This is awesome! // Positive

This is bad! // Negative

Wow that movie was rad! // Positive

What a horrible show! //

Negative

Elements of a Prompt

Instruction - a specific task or instruction you want the model to perform

Context - external information or additional context that can steer the model to better responses

Input Data - the input or question that we are interested to find a response for

Output Indicator - the type or format of the output.

Classify the text into neutral, negative, or positive

Text: I think the food was okay.

Sentiment:

Hallucinations

"An LLM hallucination occurs when a large language model (LLM) generates a response that is either factually incorrect, nonsensical, or disconnected from the input prompt." - CI Blog

by combining the emerging field of LLM evaluations (evals) with the proven techniques of continuous integration (CI), you can create a more robust system in which every update to the LLM or its training data triggers a new round of automated evals

```
Write a quiz for the category the user requests.

## Steps to create a quiz

Step 1:{delimiter} First identify the category user is asking about. Allowed categories are:

* Art * Science * Geography

If a user asks about another category respond: "I can only generate quizzes for Art, Science, or Geography"
```

Techniques

Zero-Shot Prompting

Large LLMs today, such as GPT-3, are tuned to follow instructions and are trained on large amounts of data; so they are capable of performing some tasks "zero-shot."

We tried a few zero-shot examples in the previous section. Here is one of the examples we used:

```
Classify the text into neutral,
negative or positive.

Text: I think the vacation is okay.

Output: Neutral
```

Note that in the prompt above we didn't provide the model with any examples of text alongside their classifications, the LLM already understands "sentiment" -- that's the zero-shot capabilities at work.

Few-Shot Prompting

enable **in-context learning** where we provide demonstrations in the prompt to steer the model to better performance.

The examples conditions the generated

```
Prompt: A "whatpu" is a small, furry
animal native to Tanzania. An example of
a sentence that uses the word whatpu is:

We were traveling in Africa and we saw
these very cute whatpus.

To do a "farduddle" means to jump up and
down really fast. An example of a
sentence that uses the word farduddle is:

Output: When we won the game, we all
started to farduddle in celebration.
```

Few-Shot Prompting

"the **label space and the distribution of the input text** specified by the demonstrations **are both important** (regardless of whether the labels are correct for individual inputs)"

the format you use also plays a key role in performance, **even if you just use random labels**, this is much better than no labels at all.

additional results show that **selecting random labels from a true distribution** of labels (instead of a uniform distribution) also helps. - *Min et al. (2022)*

```
Prompt: This is awesome! //
Negative

This is bad! // Positive

Wow that movie was rad! // Positive

What a horrible show! //

Output:

Negative
```

Limitations of Few-shot Prompting

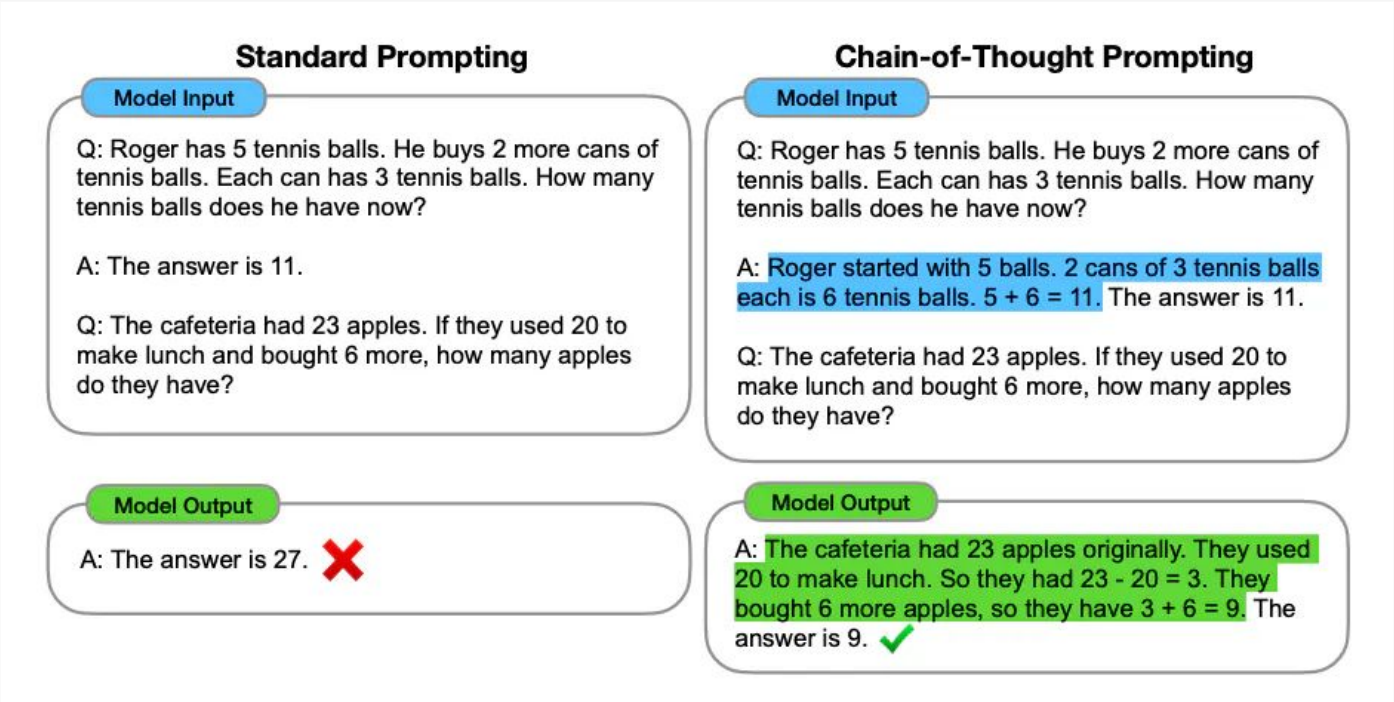
Standard few-shot prompting works well for many tasks but is still **not a perfect technique, especially when dealing with more complex reasoning tasks**. Let's demonstrate why this is the case. Do you recall the previous example where we provided the following task:

```
Prompt: The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1.

A:

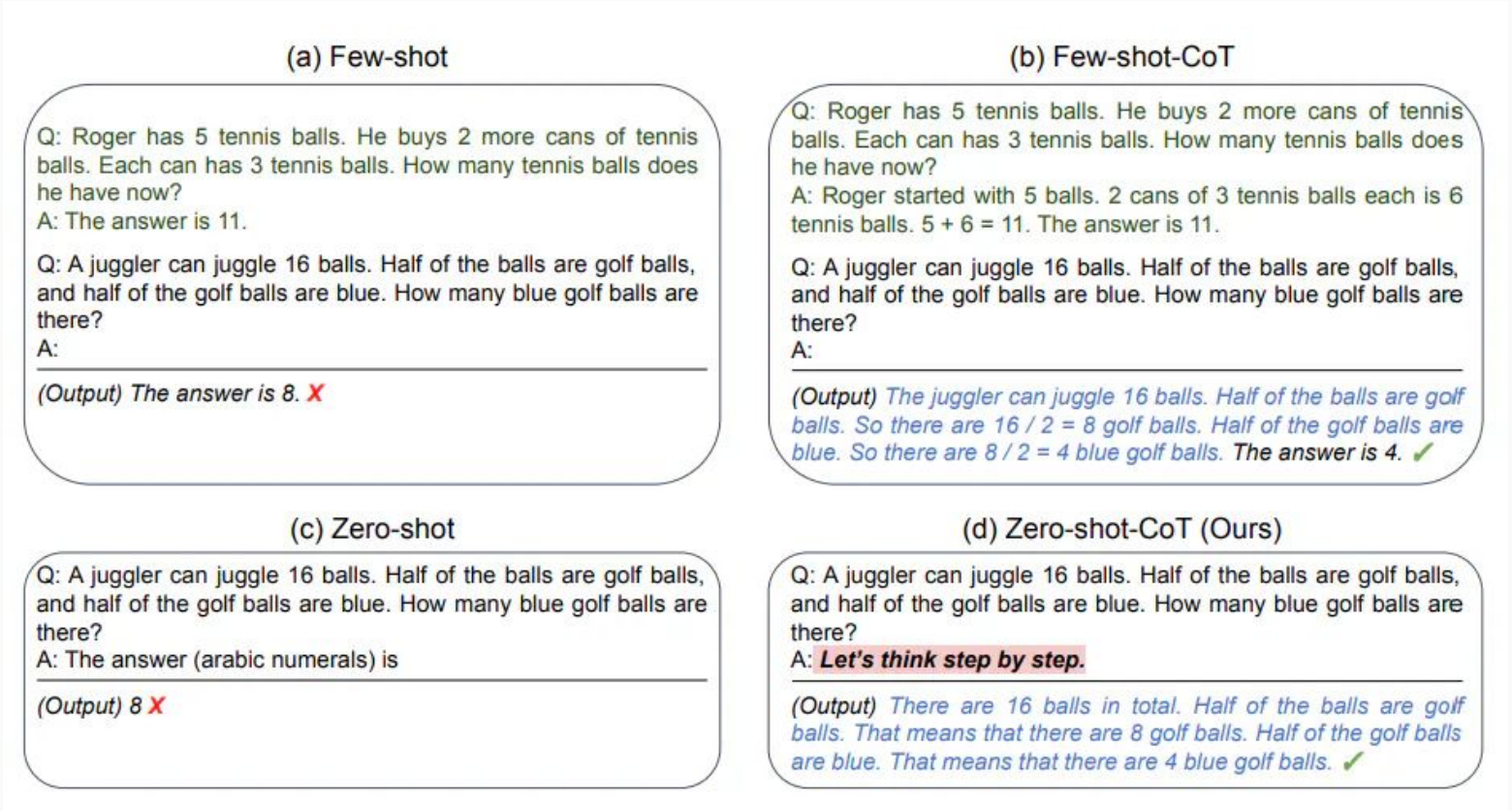
Output: Yes, the odd numbers in this group add up to 107, which is an even number.
```

Chain-of-Thought Prompting



Wei et al. (2022)

Chain-of-Thought Prompting



Automatic Chain-of-Thought (Auto-CoT)

Zhang et al. (2022) : an approach **to eliminate manual efforts** by leveraging LLMs with "Let's think step by step" prompt to generate reasoning chains for demonstrations one by one.

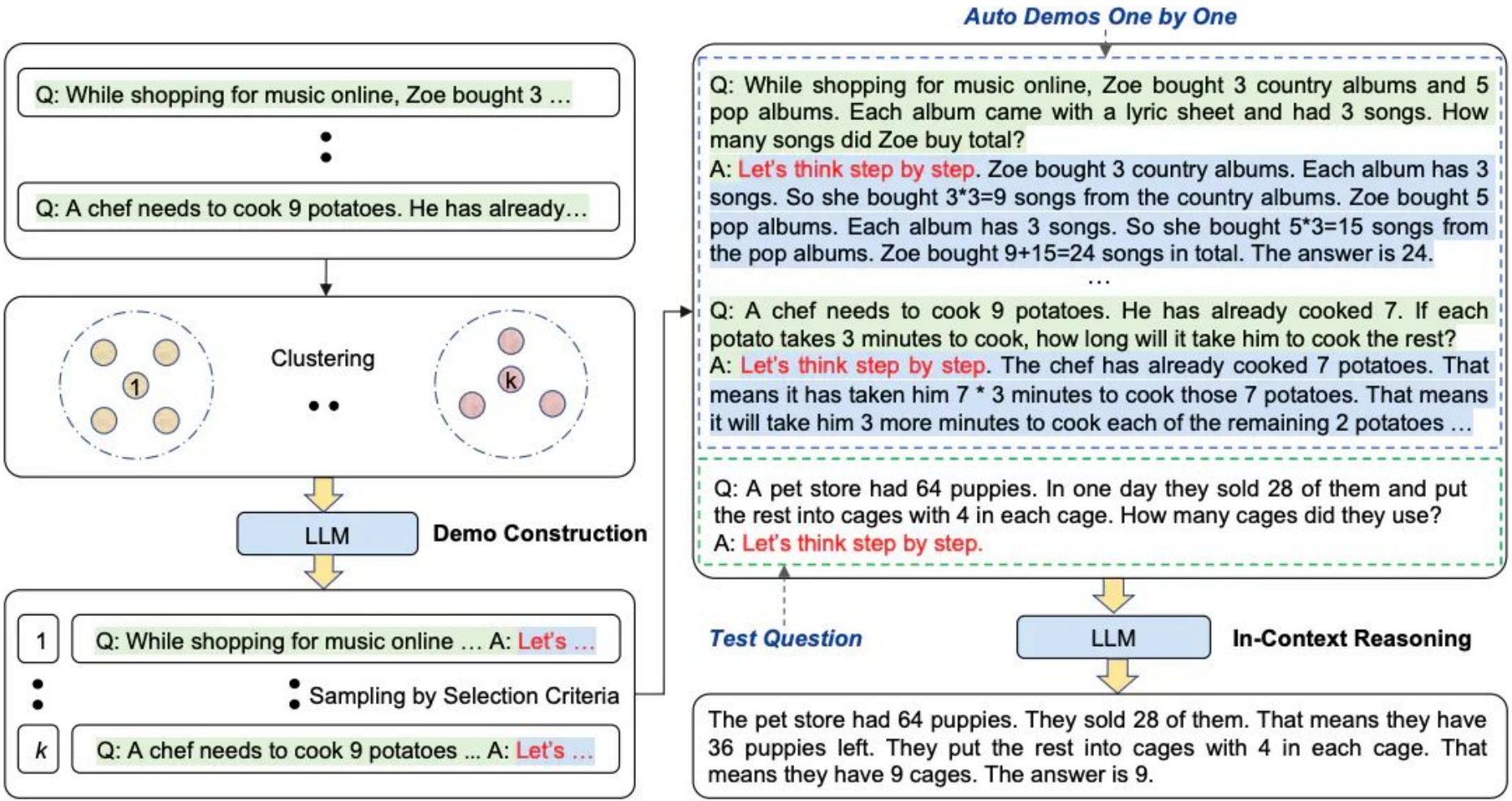
Can still end up with mistakes in generated chains. To mitigate the effects of the mistakes, the **diversity of demonstrations matter**.

GitHub - amazon-science/auto-cot: Official implementation for "Automatic Chain of Thought Prompting in LLM

This works proposes Auto-CoT, which samples questions with diversity and generates reasoning chains to construct the demonstrations.

1): question clustering: partition questions of a given dataset into a few clusters

2): demonstration sampling: select a representative question from each cluster and generate its reasoning chain using Zero-Shot-CoT with simple heuristics


Q: While shopping for music online, Zoe bought 3 …
Q: A chef needs to cook 9 potatoes. He has already…
Clustering
1
k
LLM
Demo Construction
1
Q: While shopping for music online … A: Let's …
Sampling by Selection Criteria
k
Q: A chef needs to cook 9 potatoes ... A: Let's …
Auto Demos One by One
Q: While shopping for music online, Zoe bought 3 country albums and 5 pop albums. Each album came with a lyric sheet and had 3 songs. How many songs did Zoe buy total?
A: Let's think step by step. Zoe bought 3 country albums. Each album has 3 songs. So she bought 3*3=9 songs from the country albums. Zoe bought 5 pop albums. Each album has 3 songs. So she bought 5*3=15 songs from the pop albums. Zoe bought 9+15=24 songs in total. The answer is 24.
…
Q: A chef needs to cook 9 potatoes. He has already cooked 7. If each potato takes 3 minutes to cook, how long will it take him to cook the rest?
A: Let's think step by step. The chef has already cooked 7 potatoes. That means it has taken him 7 * 3 minutes to cook those 7 potatoes. That means it will take him 3 more minutes to cook each of the remaining 2 potatoes …
Q: A pet store had 64 puppies. In one day they sold 28 of them and put the rest into cages with 4 in each cage. How many cages did they use?
A: Let's think step by step.
Test Question
LLM
In-Context Reasoning
The pet store had 64 puppies. They sold 28 of them. That means they have 36 puppies left. They put the rest into cages with 4 in each cage. That means they have 9 cages. The answer is 9.

Self-consistency

Wang et al. (2022), self-consistency aims "to replace the naive greedy decoding used in chain-of-thought prompting".

The idea is to sample multiple, diverse reasoning paths through few-shot CoT, and use the generations to select the most consistent answer.

Self-consistency

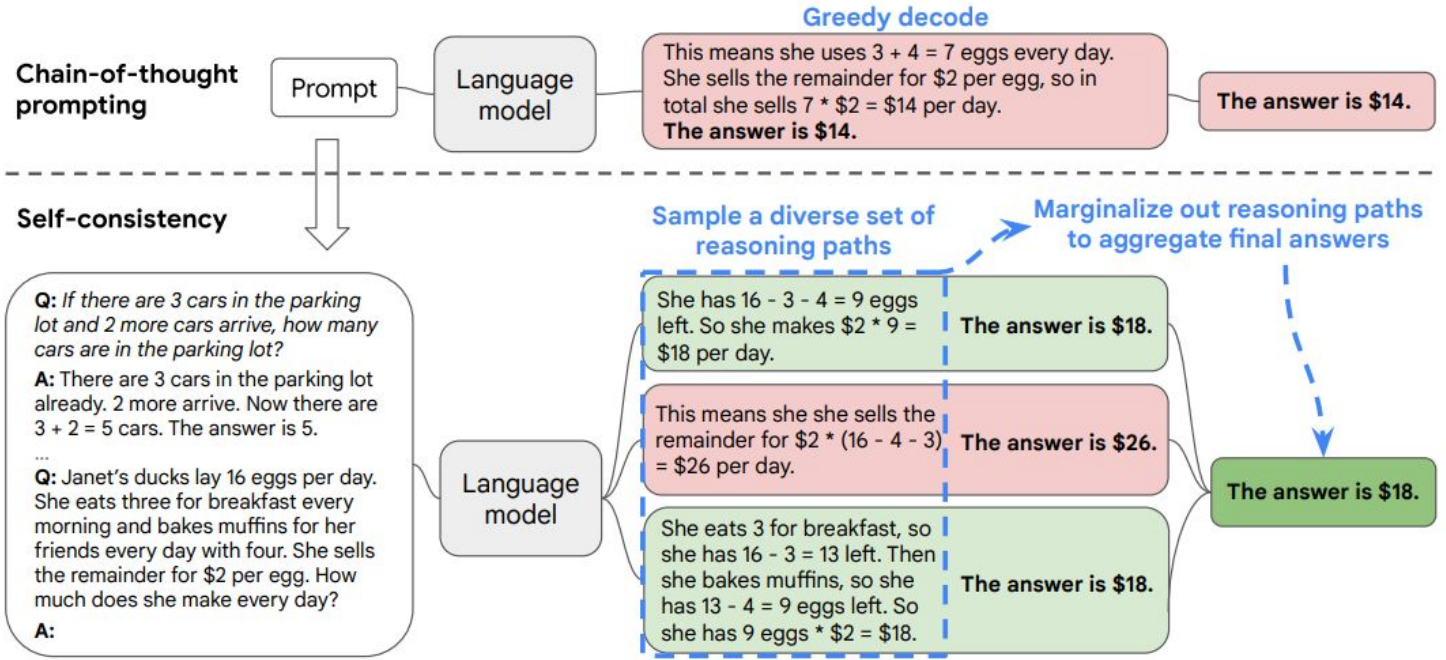


Figure 1: The self-consistency method contains three steps: (1) prompt a language model using chain-of-thought (CoT) prompting; (2) replace the "greedy decode" in CoT prompting by sampling from the language model's decoder to generate a diverse set of reasoning paths; and (3) marginalize out the reasoning paths and aggregate by choosing the most consistent answer in the final answer set.

Generated Knowledge Prompting

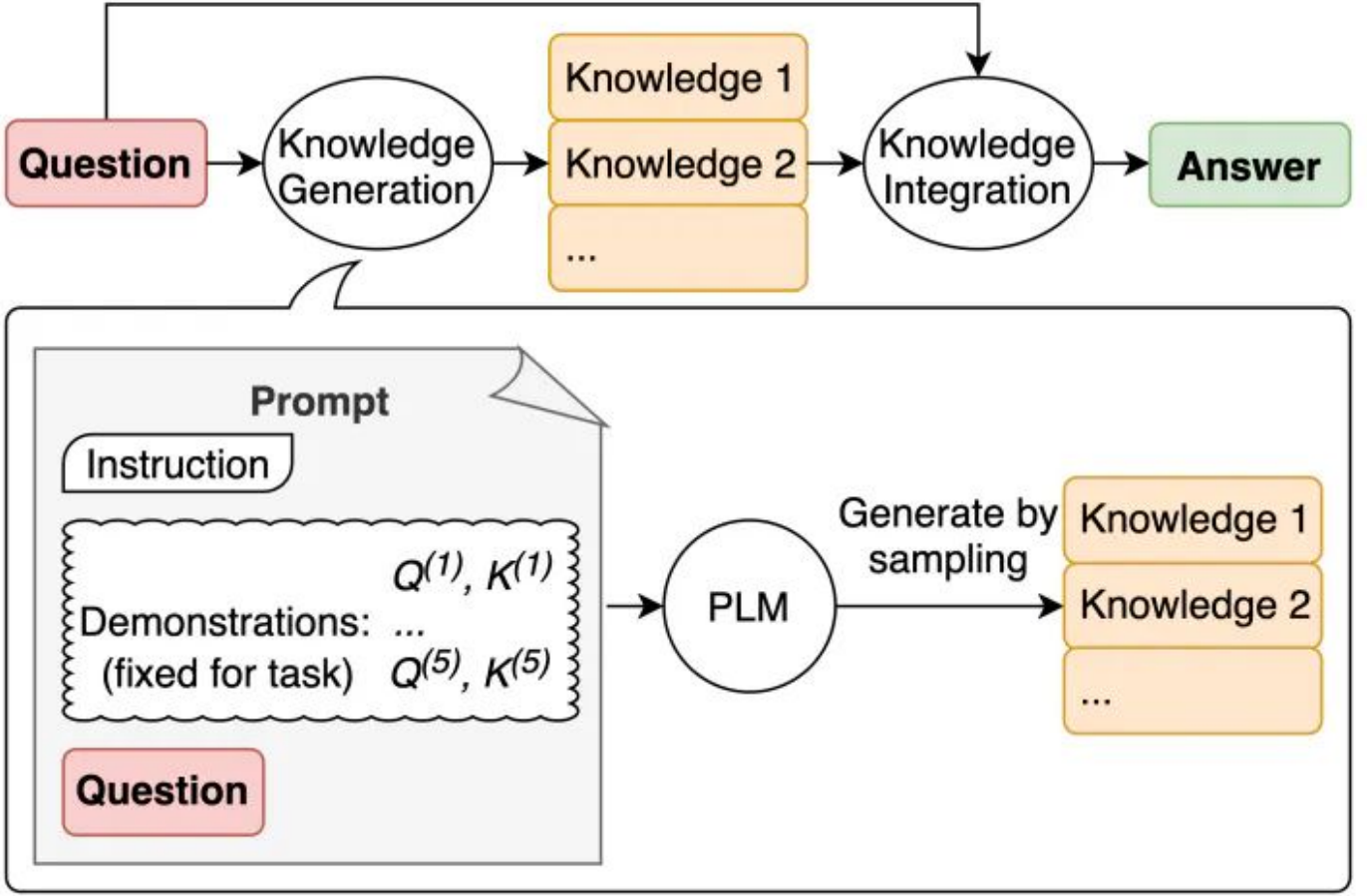
can the model also be used to generate knowledge before making a prediction? How helpful is this for tasks such as commonsense reasoning? – Liu et al. 2022

Prompt:

```
Part of golf is trying to get a higher point total
than others. Yes or No?
```

Output:

```
Yes.
```


Question
Knowledge Generation
Knowledge 1
Knowledge 2
...
Knowledge Integration
Answer
Prompt
Instruction
Demonstrations: (fixed for task)
$Q^{(1)}$, $K^{(1)}$
...
$Q^{(5)}$, $K^{(5)}$
Question
PLM
Generate by sampling
Knowledge 1
Knowledge 2
...

Generated Knowledge Prompting: 1st step

Knowledge: Those who consistently averaged less than one cigarette per day over their lifetime had nine times the risk of dying from lung cancer than never smokers. Among people who smoked between one and 10 cigarettes per day, the risk of dying from lung cancer was nearly 12 times higher than that of never smokers.

Input: A rock is the same size as a pebble.

Knowledge: A pebble is a clast of rock with a particle size of 4 to 64 millimetres based on the Udden-Wentworth scale of sedimentology. Pebbles are generally considered larger than granules (2 to 4 millimetres diameter) and smaller than cobbles (64 to 256 millimetres diameter).

Input: Part of golf is trying to get a higher point total than others.

Knowledge:

Knowledge 1: The objective of golf is to play a set of holes in the least number of strokes. A round of golf typically consists of 18 holes. Each hole is played once in the round on a standard golf course. Each stroke is counted as one point, and the total number of strokes is used to determine the winner of the game.

Knowledge 2: Golf is a precision club-and-ball sport in which competing players (or golfers) use many types of clubs to hit balls into a series of holes on a course using the fewest number of strokes. The goal is to complete the course with the lowest score, which is calculated by adding up the total number of strokes taken on each hole. The player with the lowest score wins the game.

Generated Knowledge Prompting: 2nd step

next step is to integrate the knowledge and get a prediction :

Prompt: Question: Part of golf is trying to get a higher point total than others. Yes or No?

Knowledge: The objective of golf is to play a set of holes in the least number of strokes. A round of golf typically consists of 18 holes. Each hole is played once in the round on a standard golf course. Each stroke is counted as one point, and the total number of strokes is used to determine the winner of the game.

Explain and Answer:

No, the objective of golf is not to get a higher point total than others. Rather, the objective is to play a set of holes in the least number of strokes. The total number of strokes is used to determine the winner of the game, not the total number of points.

Answer 1 (confidence very high):

Generated Knowledge Prompting: 2nd step

Question: Part of golf is trying to get a higher point total than others. Yes or No?

Knowledge: Golf is a precision club-and-ball sport in which competing players (or golfers) use many types of clubs to hit balls into a series of holes on a course using the fewest number of strokes. The goal is to complete the course with the lowest score, which is calculated by adding up the total number of strokes taken on each hole. The player with the lowest score wins the game.

Explain and Answer:

Answer 2 (confidence very low):

Answer 2 (confidence is a lot lower):

Yes, part of golf is trying to get a higher point total than others. Each player tries to complete the course with the lowest score, which is calculated by adding up the total number of strokes taken on each hole. The player with the lowest score wins the game.

Prompt chaining

Break tasks into its subtasks

LLM is prompted with a subtask and then its response is used as input to another prompt.

Useful to accomplish complex which an LLM might struggle to address if prompted with a very detailed prompt.

Helps to boost transparency of your LLM application, increases controllability, and reliability.

Prompt chaining

1. for extracting relevant quotes to answer a question and a

   ...

```
The prompting techniques mentioned in the document include:

1. Chain-of-thought (CoT) prompting[27]

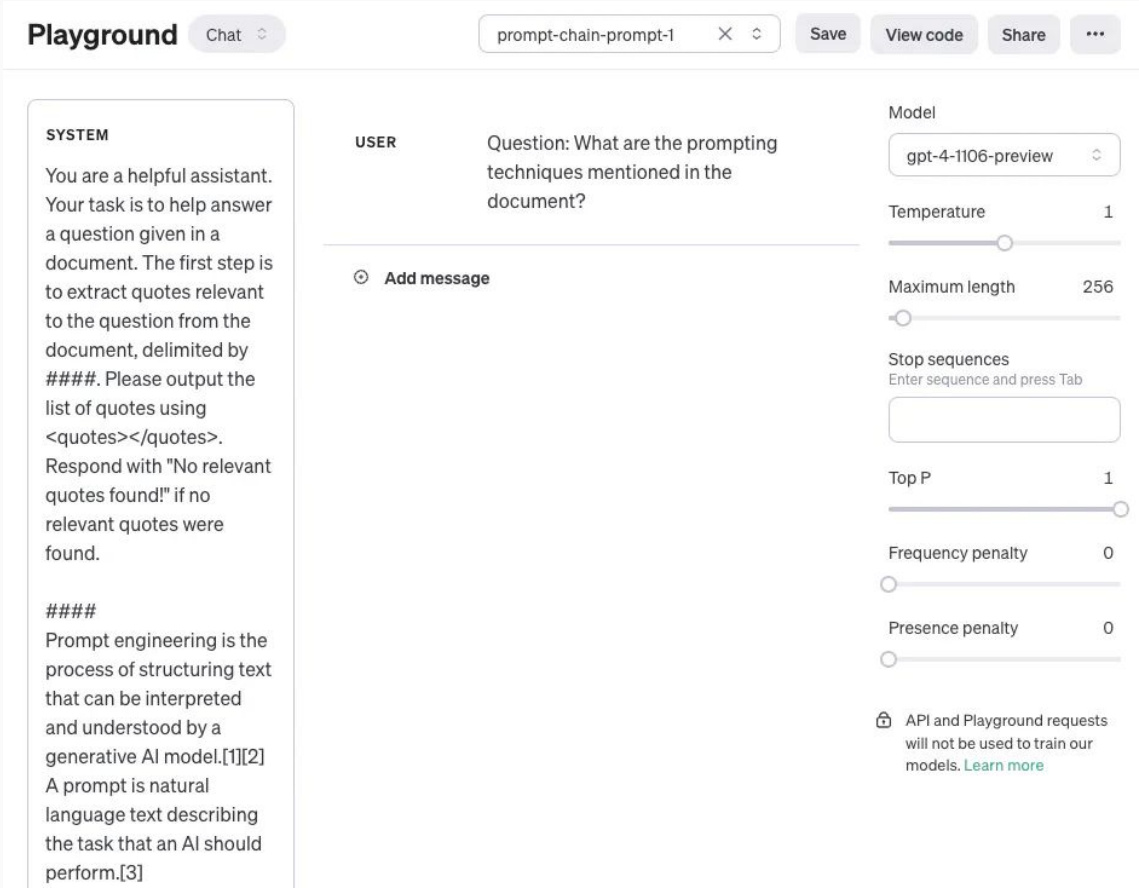
2. Generated knowledge prompting[37]

3. Least-to-most prompting[38]

4. Self-consistency decoding[39]

5. Complexity-based prompting[41]

...
```



Prompt chaining

2. takes as input the quotes and original document to answer a given question.

```
<quotes>

- Chain-of-thought (CoT) prompting[27]

- Generated knowledge prompting[37]

- Least-to-most prompting[38]

- Self-consistency decoding[39]

- Complexity-based prompting[41]

- Self-refine[42]

- Tree-of-thought prompting[43]

…
```

```
Given a set of relevant quotes (delimited by <quotes></quotes>)
extracted from a document and the original document (delimited
by ####), please compose an answer to the question. Ensure that
the answer is accurate, has a friendly tone, and sounds helpful.

####

{{document}}

####

<quotes>

- Chain-of-thought (CoT) prompting[27]

- Generated knowledge prompting[37]

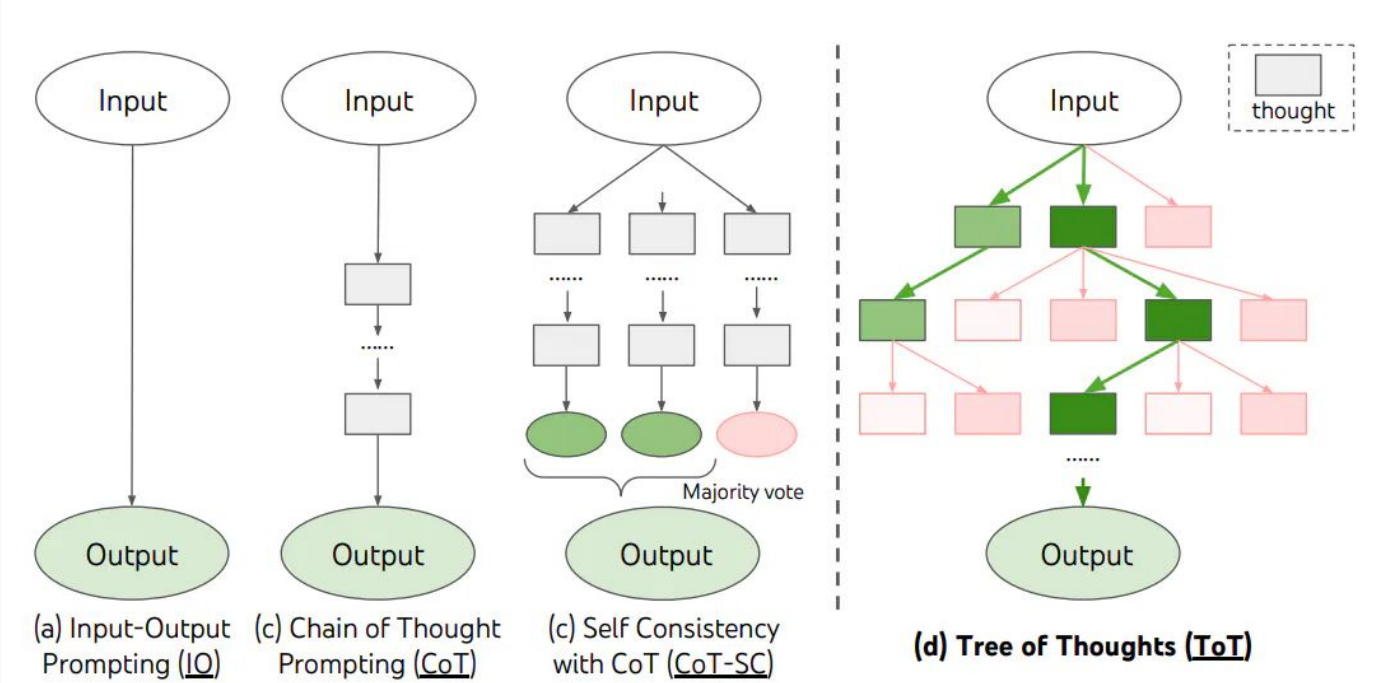
- Least-to-most prompting[38]

- Self-consistency decoding[39]

- Complexity-based prompting[41]
```

1h : https://www.deeplearning.ai/short-courses/langchain-for-llm-application-development/

Tree of thoughts



Yao et el. (2023) and Long (2023)

Tree of thoughts

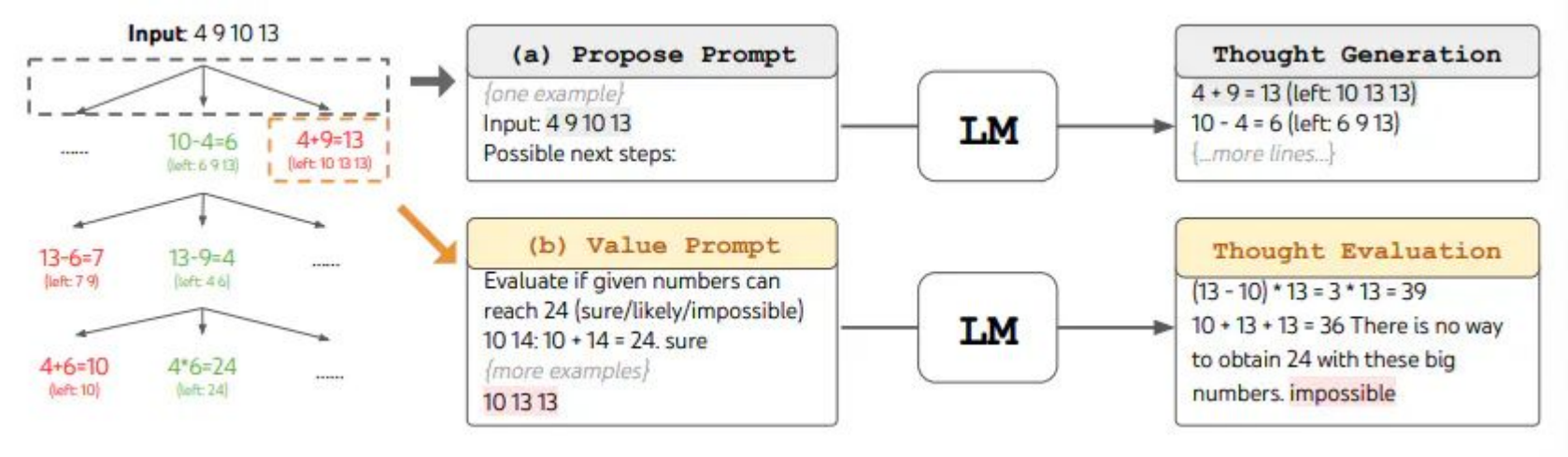
thoughts represent coherent language sequences that serve as intermediate steps toward solving a problem.

enables an LM to self-evaluate the progress intermediate thoughts make towards solving a problem through a deliberate reasoning process.

The LM's ability to generate and evaluate thoughts is then combined with search algorithms (e.g., breadth-first search and depth-first search) to enable systematic exploration of thoughts with lookahead and backtracking.

Tree of thoughts

BFS in ToT for the Game of 24 task, the LM is prompted to evaluate each thought candidate as "sure/maybe/impossible" with regard to reaching 24



"the aim is to promote correct partial solutions that can be verdicted within few lookahead trials, and eliminate impossible partial solutions based on "too big/small" commonsense, and keep the rest "maybe""

Tree of thoughts

Method	Success
IO prompt	7.3%
CoT prompt	4.0%
CoT-SC (k=100)	9.0%
ToT (ours) (b=1)	45%
ToT (ours) (b=5)	**74%**
IO + Refine (k=10)	27%
IO (best of 100)	33%
CoT (best of 100)	49%

Table 2: Game of 24 Results.

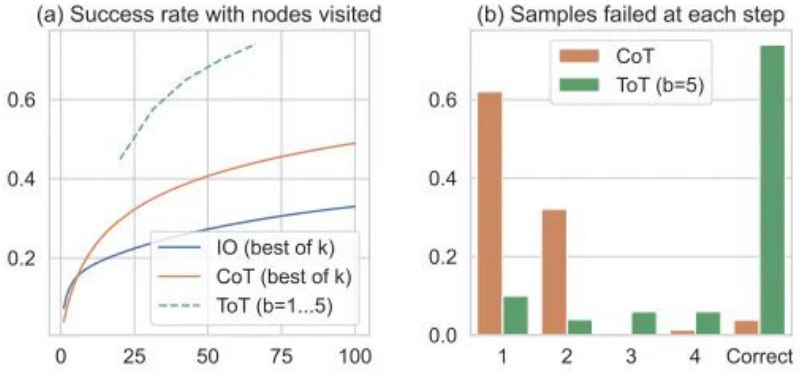


Figure 3: Game of 24 (a) scale analysis & (b) error analysis.

Code available here and here

Retrieval Augmented Generation (RAG)

For more complex and knowledge-intensive tasks, it's possible to build a language model-based system that **accesses external knowledge sources**.

This enables more factual consistency, improves reliability of the generated responses, and helps to mitigate the problem of "hallucination".

Meta AI researchers introduced a method called Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation (RAG)

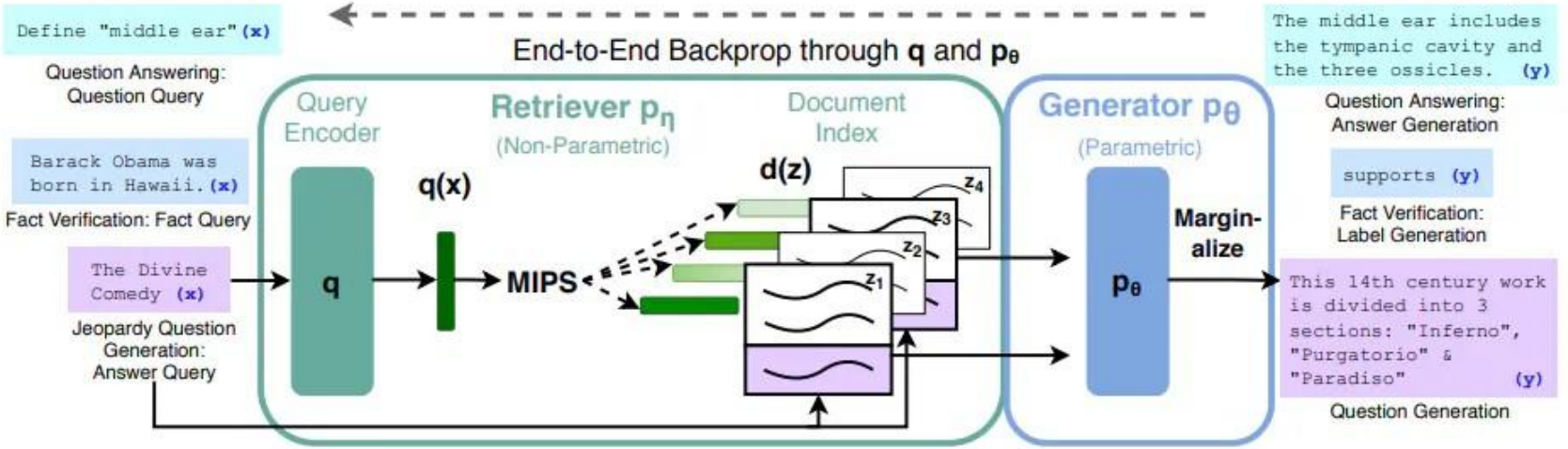


Figure 1: Overview of our approach. We combine a pre-trained retriever (*Query Encoder* + *Document Index*) with a pre-trained seq2seq model (*Generator*) and fine-tune end-to-end. For query x, we use Maximum Inner Product Search (MIPS) to find the top-K documents z_i. For final prediction y, we treat z as a latent variable and marginalize over seq2seq predictions given different documents.

Retrieval Augmented Generation (RAG)

combines an information retrieval component with a text generator model.

takes an input and retrieves a set of relevant/supporting documents given a source (e.g., Wikipedia). The documents are concatenated as context with the original input prompt and fed to the text generator which produces the final output.

RAG can be fine-tuned and its internal knowledge can be modified in an efficient manner and without needing retraining of the entire model

allows language models to bypass retraining, enabling access to the latest information for generating reliable outputs

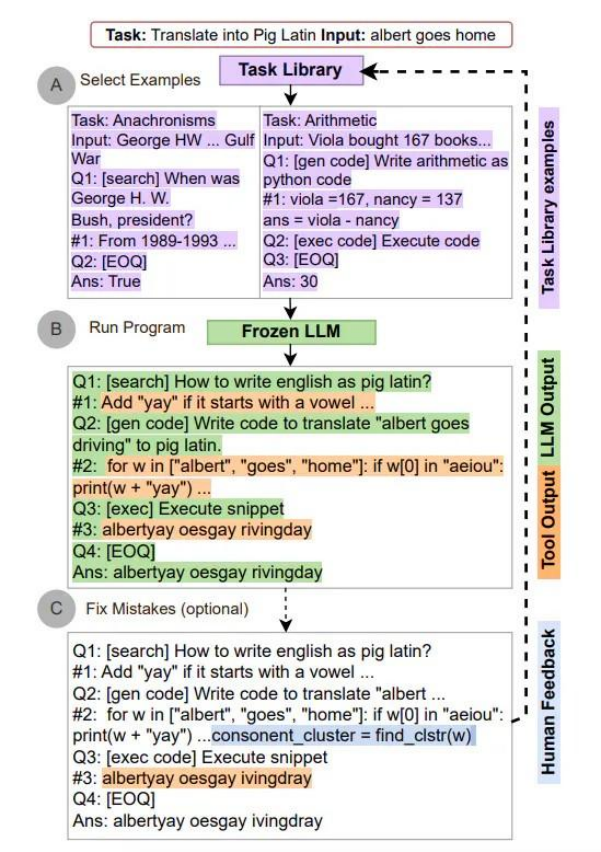
Pratique

Automatic Reasoning and Tool-use (ART)

Paranjape et al., (2023) propose a new framework that uses a frozen LLM to automatically generate intermediate reasoning steps as a program

- given a new task, it select demonstrations of multi-step reasoning and tool use from a task library
- at test time, it pauses generation whenever external tools are called, and integrate their output before resuming generation

ART also enables humans to fix mistakes in the reasoning steps or add new tools by simply updating the task and tool libraries.

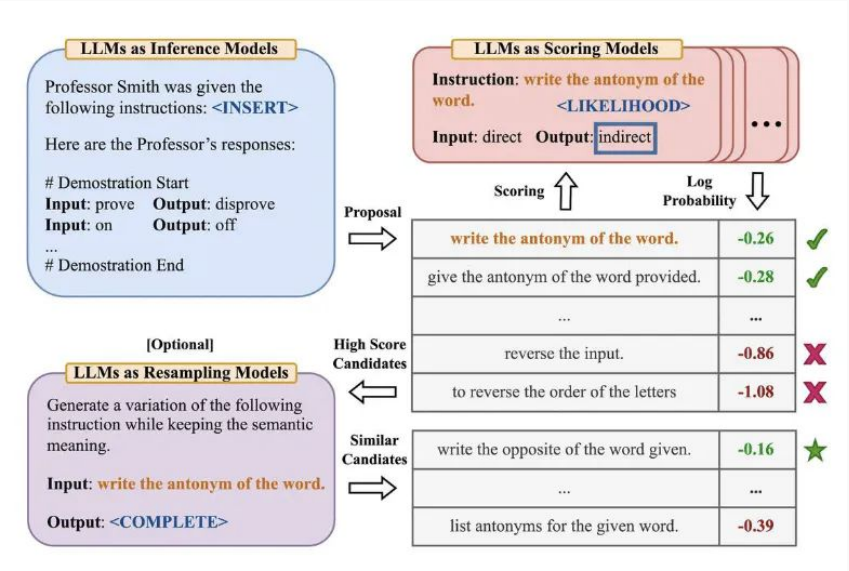

Task: Translate into Pig Latin Input: albert goes home
A
Select Examples
Task Library
Task: Anachronisms
Input: George HW ... Gulf War
Q1: [search] When was George H. W. Bush, president?
#1: From 1989-1993 ...
Q2: [EOQ]
Ans: True
Task: Arithmetic
Input: Viola bought 167 books...
Q1: [gen code] Write arithmetic as python code
#1: viola =167, nancy = 137
ans = viola - nancy
Q2: [exec code] Execute code
Q3: [EOQ]
Ans: 30
Task Library examples
B
Run Program
Frozen LLM
Q1: [search] How to write english as pig latin?
#1: Add "yay" if it starts with a vowel ...
Q2: [gen code] Write code to translate "albert goes driving" to pig latin.
#2: for w in ["albert", "goes", "home"]: if w[0] in "aeiou": print(w + "yay") ...
Q3: [exec] Execute snippet
#3: albertyay oesgay rivingday
Q4: [EOQ]
Ans: albertyay oesgay rivingday
LLM Output
Tool Output
C
Fix Mistakes (optional)
Q1: [search] How to write english as pig latin?
#1: Add "yay" if it starts with a vowel ...
Q2: [gen code] Write code to translate "albert ...
#2: for w in ["albert", "goes", "home"]: if w[0] in "aeiou": print(w + "yay") ...consonent_cluster = find_clstr(w)
Q3: [exec code] Execute snippet
#3: albertyay oesgay ivingdray
Q4: [EOQ]
Ans: albertyay oesgay ivingdray
Human Feedback

Automatic Prompt Engineer (APE)

framework for automatic instruction generation and selection

1) LLM (as an inference model) that is given **output demonstrations** to generate instruction candidates for a task
2) most appropriate instruction is selected **based on computed evaluation scores**

[Zhou et al., (2022)]



Automatic Prompt Engineer (APE)

Kojima et al., 2022 *"Let's work this out in a step by step way to be sure we have the right answer."* elicits chain-of-thought reasoning and improves performance on the MultiArith and GSM8K benchmarks:

No.	Category	Zero-shot CoT Trigger Prompt	Accuracy
1	APE	Let's work this out in a step by step way to be sure we have the right answer.	**82.0**
2	Human-Designed	Let's think step by step. (*1)	78.7
3		First, (*2)	77.3
4		Let's think about this logically.	74.5
5		Let's solve this problem by splitting it into steps. (*3)	72.2
6		Let's be realistic and think step by step.	70.8
7		Let's think like a detective step by step.	70.3
8		Let's think	57.5
9		Before we dive into the answer,	55.7
10		The answer is after the proof.	45.7
-		(Zero-shot)	17.7

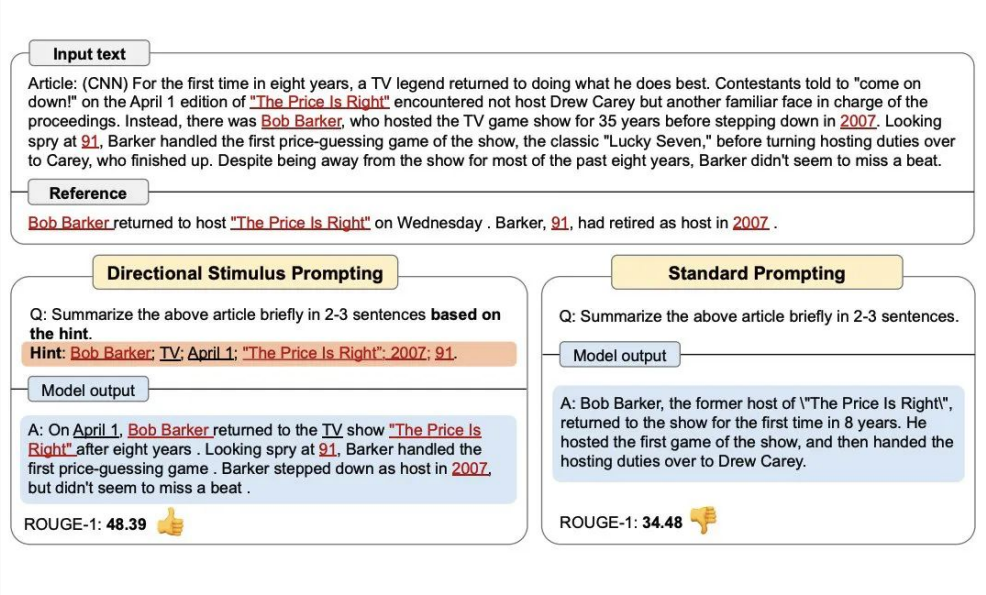
This paper touches on an important topic related to prompt engineering which is the idea of **automatically optimizing prompts**:

- Prompt-OIRL: use offline inverse reinforcement learning to generate query-dependent prompts.
- OPRO: usie LLMs to optimize prompts: let LLMs "Take a deep breath" improves the performance on math problems.
- ...

Active-Prompt

Li et al., (2023) a new prompting technique to better guide the LLM in generating the desired summary.

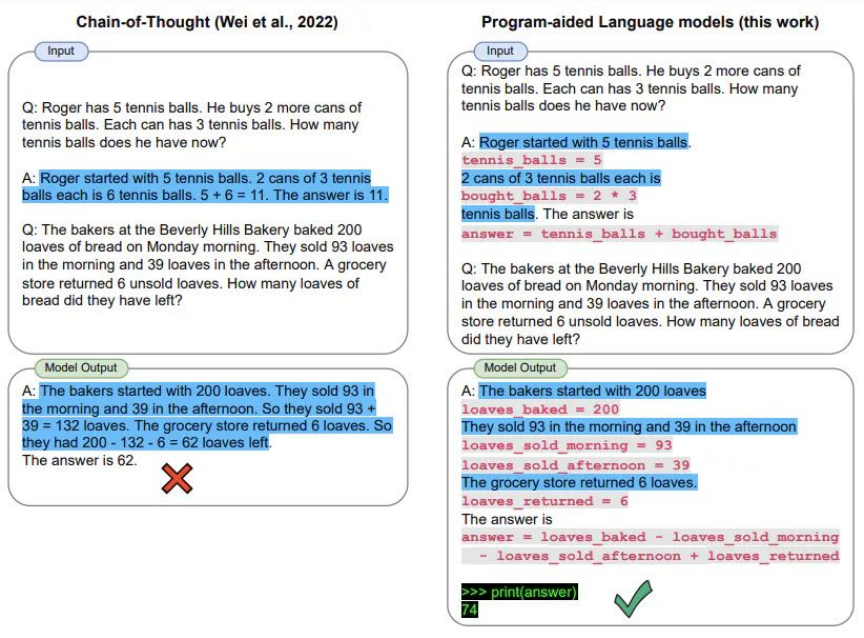
A tuneable policy LM is trained to generate the stimulus/hint. Seeing more use of RL to optimize LLMs.



PAL (Program-Aided Language Models)

Gao et al., (2022) read natural language problems and generate programs as the intermediate reasoning steps.

differs from chain-of-thought prompting in that instead of using free-form text to obtain solution it offloads the solution step to a programmatic runtime such as a Python interpreter.



PAL (Program-Aided Language Models)

Test !

PAL (Program-Aided Language Models)

```
import openai
from datetime import datetime
from dateutil.relativedelta import relativedelta
import os
from langchain.llms import OpenAI
from dotenv import load_dotenv
```

Let's first configure a few things:

```
load_dotenv()

# API configuration
openai.api_key = os.getenv("OPENAI_API_KEY")

# for LangChain
os.environ["OPENAI_API_KEY"] = os.getenv("OPENAI_API_KEY")
```

Setup model instance:

```
llm = OpenAI(model_name='text-davinci-003', temperature=0)
```

Setup prompt + question:

```
question = "Today is 27 February 2023. I was born exactly 25 years ago. What is the date I was born in MM/DD/YYYY?"
```

```
DATE_UNDERSTANDING_PROMPT = """
# Q: 2015 is coming in 36 hours. What is the date one week from today in MM/DD/YYYY?
# If 2015 is coming in 36 hours, then today is 36 hours before.
today = datetime(2015, 1, 1) - relativedelta(hours=36)
# One week from today,
one_week_from_today = today + relativedelta(weeks=1)
# The answer formatted with %m/%d/%Y is
one_week_from_today.strftime('%m/%d/%Y')
# Q: The first day of 2019 is a Tuesday, and today is the first Monday of 2019. What is the date today in MM/DD/YYYY?
# If the first day of 2019 is a Tuesday, and today is the first Monday of 2019, then today is 6 days later.
today = datetime(2019, 1, 1) + relativedelta(days=6)
# The answer formatted with %m/%d/%Y is
today.strftime('%m/%d/%Y')
# Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?
# If the concert was scheduled to be on 06/01/1943, but was delayed by one day to today, then today is one day later.
today = datetime(1943, 6, 1) + relativedelta(days=1)
# 10 days ago,
ten_days_ago = today - relativedelta(days=10)
# The answer formatted with %m/%d/%Y is
ten_days_ago.strftime('%m/%d/%Y')
# Q: It is 4/19/1969 today. What is the date 24 hours later in MM/DD/YYYY?
# It is 4/19/1969 today.
today = datetime(1969, 4, 19)
# 24 hours later,
later = today + relativedelta(hours=24)
# The answer formatted with %m/%d/%Y is
today.strftime('%m/%d/%Y')
# Q: Jane thought today is 3/11/2002, but today is in fact Mar 12, which is 1 day later. What is the date 24 hours later in MM/DD/YYYY?
# If Jane thought today is 3/11/2002, but today is in fact Mar 12, then today is 3/12/2002.
today = datetime(2002, 3, 12)
# 24 hours later,
later = today + relativedelta(hours=24)
# The answer formatted with %m/%d/%Y is
```

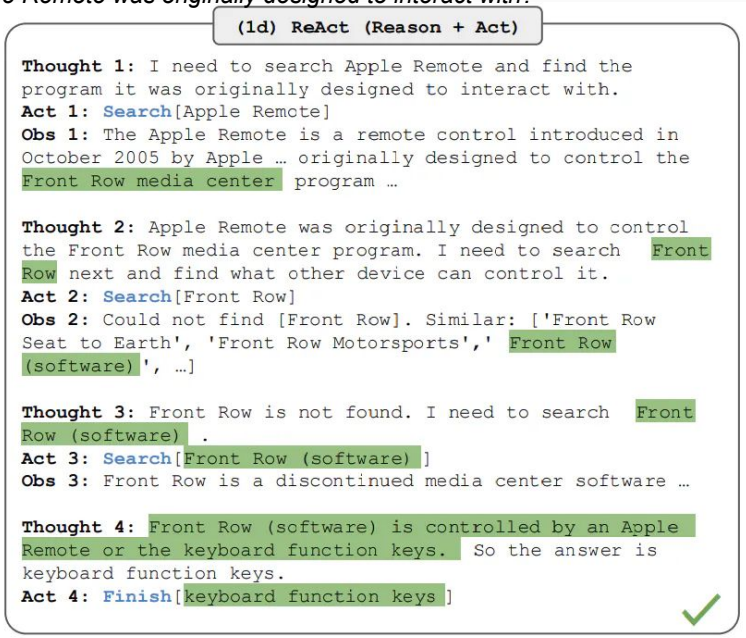
ReAct Prompting

a framework to generate both **reasoning traces** and **task-specific actions** in an interleaved manner.

- perform dynamic reasoning to create, maintain, and adjust plans for acting
- while also enabling interaction to external environments (e.g., Wikipedia) to incorporate additional information into the reasoning.

[Yao et al., 2022]

Aside from the Apple Remote, what other devices can control the program Apple Remote was originally designed to interact with?



Automatic Prompt Engineer (APE)

Results on Knowledge-Intensive Tasks

ReAct outperforms CoT on Fever

and lags behind CoT on HotpotQA.

A detailed error analysis is provided:

- CoT suffers from **fact hallucination**
- ReAct's structural constraint reduces its flexibility in formulating reasoning steps
- ReAct depends a lot on the information it's retrieving; **non-informative search results derails the model reasoning** and leads to difficulty in recovering and reformulating thoughts

Prompt Method[a]	**HotpotQA** (EM)	**Fever** (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC → ReAct	34.2	**64.6**
ReAct → CoT-SC	**35.1**	62.0
Supervised SoTA[b]	67.5	89.5

ReAct Prompting

Test !

Verification

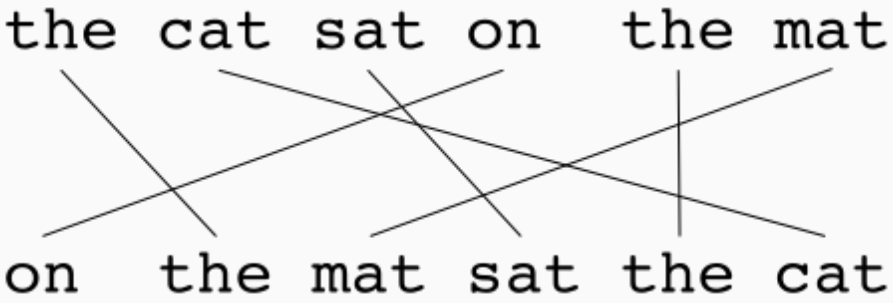
Métriques

BLEU — Measures precision of word n-grams between generated and reference texts.

ROUGE — Measures recall of word n-grams and longest common sequences.

METEOR — Incorporates recall, precision, and additional semantic matching based on stems and paraphrasing.

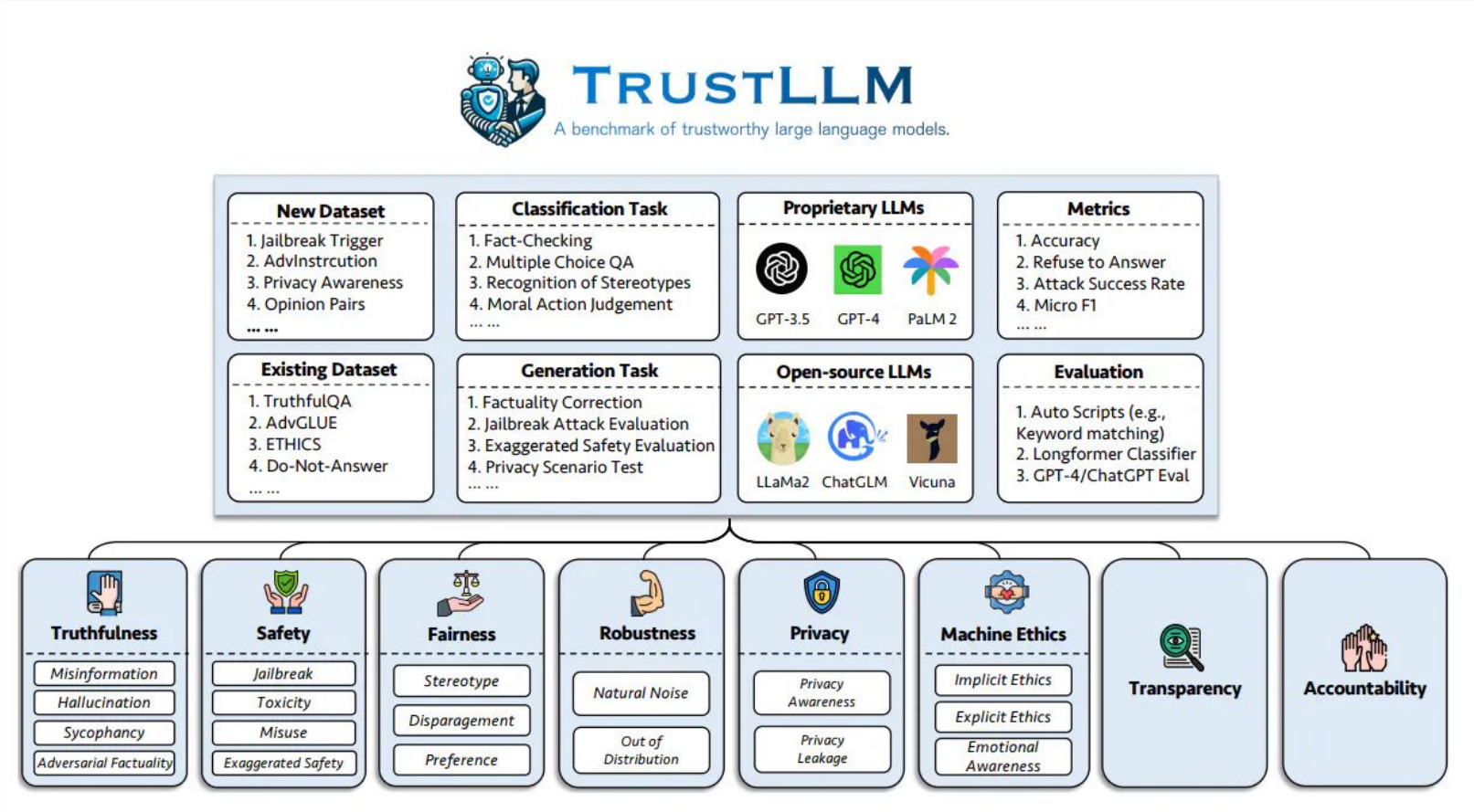
BERTScore — Matches words/phrases using BERT contextual embeddings and provides token-level granularity.



Metrique

Metric	Use Case	Method	Why to Use
ROUGE	Text Summarization	Measures overlap of N-grams and Longest Common Subsequence (LCS) between summaries	Popular for summarization tasks; captures content overlap between system-generated and reference summaries
BLEU	Machine Translation	Measures N-gram precision between candidate and reference translations	Most popular for translation tasks; captures word-by-word similarity
METEOR	Machine Translation, Text Generation	Calculates harmonic mean of unigram precision and recall, with a penalty for length mismatches	Can be used for various text generation tasks; balances precision and recall
BERTScore	Text Summarization, Machine Translation, Text Similarity	Computes cosine similarity between contextualized embeddings of words in sentences	Captures semantic similarity between sentences; applicable to various NLP tasks

Confiance


TRUSTLLM
A benchmark of trustworthy large language models.
New Dataset
1. Jailbreak Trigger
2. AdvInstrcution
3. Privacy Awareness
4. Opinion Pairs
... ...
Classification Task
1. Fact-Checking
2. Multiple Choice QA
3. Recognition of Stereotypes
4. Moral Action Judgement
... ...
Proprietary LLMs
GPT-3.5
GPT-4
PaLM 2
Metrics
1. Accuracy
2. Refuse to Answer
3. Attack Success Rate
4. Micro F1
... ...
Existing Dataset
1. TruthfulQA
2. AdvGLUE
3. ETHICS
4. Do-Not-Answer
... ...
Generation Task
1. Factuality Correction
2. Jailbreak Attack Evaluation
3. Exaggerated Safety Evaluation
4. Privacy Scenario Test
... ...
Open-source LLMs
LLaMa2
ChatGLM
Vicuna
Evaluation
1. Auto Scripts (e.g., Keyword matching)
2. Longformer Classifier
3. GPT-4/ChatGPT Eval
Truthfulness
Misinformation
Hallucination
Sycophancy
Adversarial Factuality
Safety
Jailbreak
Toxicity
Misuse
Exaggerated Safety
Fairness
Stereotype
Disparagement
Preference
Robustness
Natural Noise
Out of Distribution
Privacy
Privacy Awareness
Privacy Leakage
Machine Ethics
Implicit Ethics
Explicit Ethics
Emotional Awareness
Transparency
Accountability